



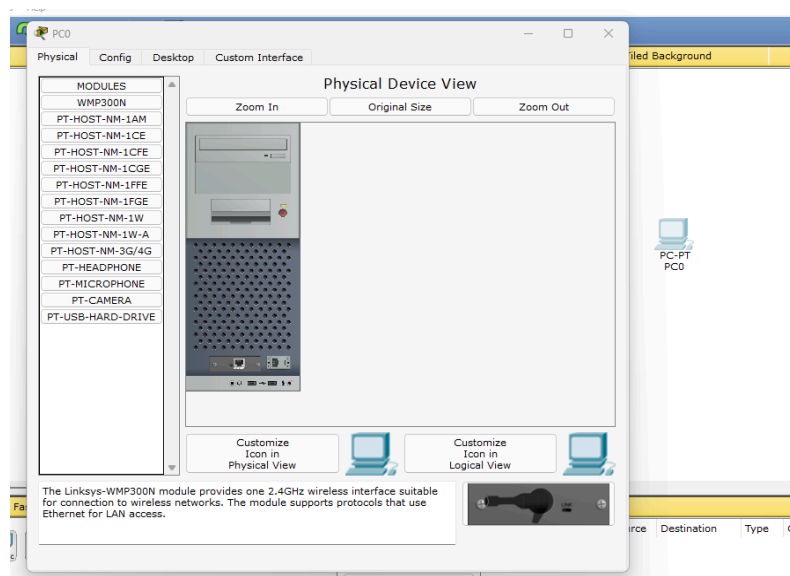
## Lab 2: Network Hardware, Topologies and Connections

### Introduction to Network Hardware

Network hardware consists of the physical devices that are essential for communication and interaction between devices in a computer network. Some key components include:

1. **Routers**, which forward data packets between different networks and are commonly used to connect a local network to the internet.
2. **Switches**, which interconnect multiple devices in the same network and ensure that data packets are forwarded to the correct device using their MAC addresses.
3. **Hubs**, which are basic devices used to connect multiple devices in a local area network (LAN), although they broadcast data to all devices, which can lead to congestion.
4. **Network Interface Cards (NICs)**, which are hardware components installed in computers and other devices to enable network connections.
5. **Cables and Connectors**, which serve as the physical medium for connecting devices. Examples include **Ethernet cables** and **fiber optic cables**, which provide high-speed data transmission.

All of these hardware components can be modelled and observed in CISCO. If you go to CISCO, drag a device to your logical workspace and then click on it, you will be able to partially observe a physical representation of the device you clicked on. An example of a PC's physical representation can be observed in the image below:

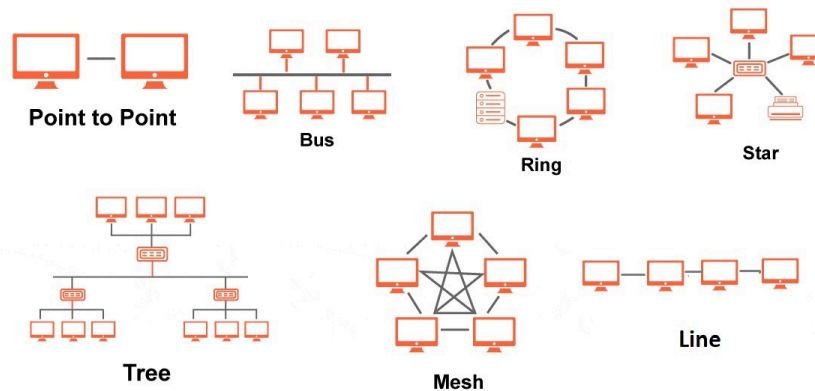




## Introduction to Network Topologies

A network topology refers to the arrangement of various elements (such as links and nodes) in a computer network. It defines how different devices are interconnected and how data flows between them. Understanding different network topologies is critical for designing networks that are efficient, scalable, and reliable. Common network topologies include **star**, **ring**, **bus**, and **mesh**. Each topology comes with its own unique advantages and disadvantages, which can impact network performance, reliability, and ease of troubleshooting.

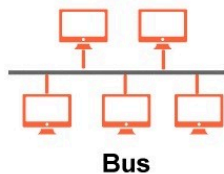
### Network Topology Types



### Examples of network topologies

Let's delve a little deeper into the different topologies and how they work. We will first start with the bus topology.

## Bus Topology

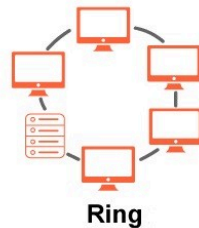


In a bus topology, all devices, or nodes, are connected to a single central cable called the "bus". This cable acts as the shared communication medium, and data is transmitted along it in both directions. Each device on the bus listens for data addressed specifically to it while ignoring other communications. At both ends of the bus, terminators are installed to absorb signals and



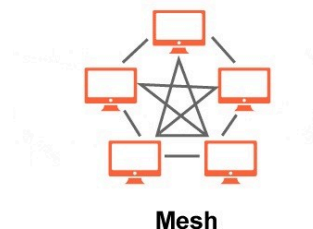
prevent data reflection, which can cause interference and disrupt the network. If the central cable fails, the entire network is disrupted. This topology is cost-effective and simple to implement but is best suited for small networks with limited devices, as performance deteriorates with increased traffic or devices.

## Ring Topology



In a ring topology, devices are connected in a closed-loop structure, where each device is linked to two others, forming a circular network. Data travels around the ring in a unidirectional manner. A token-passing mechanism is often used, ensuring that only one device can send data at a time to avoid collisions. Each device in the ring acts as a repeater, forwarding data to the next device until it reaches its intended destination. While this topology is relatively simple and prevents data collisions, a single device or link failure can disrupt the entire network unless fault tolerance mechanisms like dual rings (where data travels bidirectionally) are implemented. Ring topology was historically used in legacy systems, such as token ring networks in industrial or office settings.

## Mesh Topology

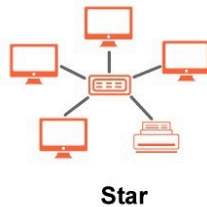


Mesh topology provides a highly interconnected network structure where every device is directly connected to every other device. This topology can be categorized into full mesh, where all devices have direct links, and partial mesh, where only some devices are directly connected, with the rest communicating indirectly. Data in a mesh topology can take multiple paths to reach its destination, allowing the network to dynamically select the most efficient route. This ensures high reliability and fault tolerance, as the network remains operational even if one or more links fail. However, the complexity and cost of installation increase significantly with the number of



devices due to the extensive cabling and connections required. Mesh topology is often used in environments that demand high availability and reliability, such as military communications, financial networks, and IoT systems.

## Star Topology



In a star topology, all devices in the network are connected to a central hub, switch, or router, which acts as a communication point. Data from any device is first sent to the central hub, which then forwards it to the intended recipient. This design ensures that the failure of one device does not affect the rest of the network. However, the central hub itself becomes a single point of failure; if it fails, the entire network is disrupted. Star topology is easy to set up and manage, making it one of the most commonly used designs in modern networks. It is frequently employed in home and office environments, typically using Ethernet switches or Wi-Fi routers for connectivity.

As you can see, there are many different ways in which you can arrange the devices on your network. However, their arrangement is not the only thing you can configure. You can also configure *the way* devices connect to each other on the network. There is a simple dichotomy to this: wired and wireless. You have already studied how to connect devices through wires so let's take a look at how you can achieve the same without wires.

## Setting Up a Wireless Connection



To create a wireless network in Cisco Packet Tracer, start by setting up the central device, a **WRT300N wireless router**. This router is ideal for basic wireless configurations and supports the required settings for this lab. Drag the WRT300N from the Wireless Devices section onto the workspace. Once placed, click on the router and navigate to the **GUI tab**. Here, disable the



DHCP server to manually assign static IP addresses, which provides better control over the network configuration.

Next, go to the **Wireless tab** and rename the SSID (network name) to *labtwo*, ensuring it is easily identifiable. Save these settings, then move to the **Wireless Security tab**. Select **WPA2 Personal** as the security protocol for its strong encryption and set a network password, such as *secure1234*, to ensure only authorized devices can connect. Save the security settings to complete the router setup.

With the router configured, bring two PCs onto the workspace from the End Devices section. For each PC, the wireless capability must be enabled. Click on the first PC, open the **Physical tab**, and turn off the device by pressing the power button. Remove the default Ethernet module and replace it with the **WMP300N wireless module**, which allows the PC to connect to the wireless network. Once the module is in place, power the PC back on.

Now, configure the wireless settings on the PC. Navigate to the **Desktop tab**, open **PC Wireless**, and scan for available networks. Select the network *labtwo*, enter the password *secure1234*, and connect. Upon successful connection, the message "Successfully connected to the access point" will appear.

After connecting, go to **IP Configuration** on the PC. Select the **Static** option and assign the PC an IP address, such as *192.168.0.5*. Set the default gateway to the router's IP address, typically *192.168.0.1*. This gateway configuration ensures communication within the network.

Repeat the same process for the second PC, replacing the IP address with *192.168.0.6*. Once both PCs are connected to the wireless network and have their IP configurations set, test the connection between them. Open the **Command Prompt** on the first PC and use the *ping* command to send packets to the second PC's IP address (*192.168.0.6*). If the connection is successful, you will see replies from the second PC, confirming that the wireless network is operational.

By following these steps, you will have successfully created and tested a wireless connection in Cisco Packet Tracer, allowing the two PCs to communicate wirelessly through the WRT300N router.



## Lab Tasks

**Note: All tasks must be documented in a single report. Include screenshots of every step (description is not as necessary as the screenshots).**

**CLO2: Apply core networking concepts and protocols, including but not limited to routing, Congestion control, and error management, to establish and analyze reliable data communication across networks.**

### Task 1: Network Hardware Exploration (10 Marks)

For this task, you will open Cisco Packet Tracer and add several devices to your workspace. Begin by adding the following devices:

- One **1841 Router** to serve as the central routing device for interconnecting networks.
- One **2960-24TT Switch** to create the central hub for connecting devices within the same network.
- One **Generic Hub** to demonstrate a simple LAN connection where all devices are connected to a single device that broadcasts data to all.
- Two **PC-PT** devices, which will represent end-user computers in the network, capable of sending and receiving data.
- One **Laptop-PT** for wireless connectivity to the network, representing a portable device.
- One **Wireless Router-PT** to act as the source of the wireless network, connecting wireless devices to the local network.

Once the devices are added, each device should be labeled with its name and a brief description of its purpose. For example, the **router** connects different networks, while the **switch** connects devices within the same network. Take a screenshot of the entire setup and ensure each device is clearly labeled with its name and function (one line is enough).

Next, watch these two videos related to Ethernet cable creation and wiring standards:

1. Ethernet Cables, UTP vs STP, Straight vs Crossover:  
[▶ Ethernet Cables, UTP vs STP, Straight vs Crossover, CAT 5,5e,6,7,8 Network Cab...](#)
2. How to Wire Up Ethernet Plugs:  
[▶ How to Wire Up Ethernet Plugs the EASY WAY! \(Cat5e / Cat6 RJ45 Pass Through...](#)

Take a screenshot of your browser with a notepad/sticky note open listing your name, roll number, section and lab number for **each** of the videos and include it in your report. Then, based on the videos, answer the following questions in your report. Keep your answers concise (1-2 lines are enough).

1. What is the function of an RJ45 connector?



2. What is the primary difference between the T568A and T568B wiring standards?
3. How does the wiring of a "Crossover" cable differ from a "Straight-through" cable?
4. When connecting a computer directly to another computer (without a switch), which cable type should be used?
5. List the correct color order for the T568B standard (from pin 1 to 8).

## Task 2: Create and Simulate Different Network Topologies (10 Marks)

For this task, you will be required to simulate four different network topologies using Cisco Packet Tracer. Each topology should be configured with **four PC-PT** devices and an appropriate network structure. For each topology, assign IP addresses to the PCs and simulate packet transmission using the **ping** command to test connectivity.

1. **Star Topology:** Create a **star topology**, where all four PCs are connected to a central **2960-24TT Switch**. Assign IP addresses to each PC, such as:
  - PC1: 192.168.1.1
  - PC2: 192.168.1.2
  - PC3: 192.168.1.3
  - PC4: 192.168.1.4
  - After assigning the IP addresses, use the **ping** command to simulate communication between the PCs.

In your documentation, provide one advantage or disadvantage of the star topology.

2. **Ring Topology:** Now, create a **ring topology** where the four PCs are connected in a circular fashion (**hint:** you may need to use 4 switches). Assign IP addresses to each PC as follows:
  - PC1: 192.168.2.1
  - PC2: 192.168.2.2
  - PC3: 192.168.2.3
  - PC4: 192.168.2.4
  - Use the **ping** command to simulate packet transmission and observe the flow of data.

In your documentation, provide one advantage or disadvantage of the ring topology.

3. **Bus Topology:** Next, create a **bus topology** with 4 devices (**hint:** you may need 4 switches). Assign the following IP addresses:
  - PC1: 192.168.3.1
  - PC2: 192.168.3.2
  - PC3: 192.168.3.3
  - PC4: 192.168.3.4



- After assigning IP addresses, use the **ping** command to check the connectivity between the devices.

In your documentation, provide one advantage or disadvantage of the bus topology.

4. **Mesh Topology:** Finally, create a **mesh topology** where each of the four PCs is connected to every other device in the network (**hint:** you may need some switches).

Assign the IP addresses:

- PC1: 192.168.4.1
- PC2: 192.168.4.2
- PC3: 192.168.4.3
- PC4: 192.168.4.4
- Use the **ping** command to test the connectivity between the PCs.

In your documentation, provide one advantage or disadvantage of the mesh topology.

**Note:** To understand the advantages/disadvantages of each topology a bit better, you can watch [this video](#).

### Task 3: Configure a Wireless Network with 3 PCs (5 Marks)

In this task, you will configure a wireless network using Cisco Packet Tracer, involving a WRT300N wireless router and three PCs. Start by placing the router and PCs in the workspace, ensuring the router is centrally located with the PCs positioned within its wireless range. Assign static IP addresses to the PCs: 192.168.0.10 for PC1, 192.168.0.11 for PC2, and 192.168.0.12 for PC3, with the subnet mask set to 255.255.255.0.

Next, access the WRT300N router's settings and configure the SSID to "*cnet\_lab2*" and enable WPA2-Personal encryption with the password as **your roll number** for security. Save the configuration and test connectivity between the PCs using the ping command to verify successful communication.

Finally, take a screenshot of the network setup showing the wireless connections and IP addresses, and write a brief description outlining the configuration process, including the SSID, encryption settings, and IP address assignments.

### Submission Guidelines (5 Marks):

- Submit a single PDF document containing all tasks and the CISCO packet file(s) to the GCR, **ensure your file name is visible in all pictures and contains your roll number**.
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter\_RollNumber\_LabNumber, for example: **L\_i231234\_Lab1**.





## CL3001 - Computer Networks Lab

Instructor: Ms. Zoya Mahboob

### Lab 2 - Network Hardware, Topologies and Connections

- Name the CISCO files as follows: SectionLetter\_RollNumber\_TaskNumber, for example: **L\_i231234\_Lab1**.
- The unethical usage of Generative AI is strictly prohibited and will result in 0 marks.
- In case a demo is conducted, there will be deductions based on your answers to the questions asked.

**NOTE: Screenshots where your name, roll number, section AND lab number are NOT visible, will automatically be given 0/not considered valid.**

Sample screenshot:

